

MirageBeam: Stealthy Adversarial Beamforming Feedback for Resilient WiFi Sensing Hijacking

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Abstract—Beamforming feedback information (BFI) sensing is emerging as a promising successor to channel state information (CSI)-based WiFi sensing thanks to its accessibility on commodity hardware without driver hacking. Yet, its security implications have been scarcely studied. Unlike CSI, BFI is broadcast in plaintext, offering adversaries a broader attack surface. We present *MirageBeam*, the first practical framework for adversarial attacks on BFI-based sensing. By injecting carefully crafted BFI frames, adversaries can launch both untargeted and targeted attacks that cause misclassification in behavior recognition systems. The core challenge is to maximize misclassification while remaining indistinguishable from regular behavior samples. We formulate an optimization problem that jointly selects the forged BFI amplitude and injection timing to covertly control the BFI sequence observed by the receiver. Furthermore, real-world injections face the challenges of attack asynchrony caused by unpredictable injection delays. We introduce a delayed-BFI injection window and recast the optimization as a weighted formulation over delay units, enabling seamless adaptation to varying delays. Implemented on commodity WiFi hardware across three indoor settings, *MirageBeam* achieves over 81% attack success while maintaining a detection rate below 30%, revealing critical vulnerabilities in existing BFI sensing systems. Actionable countermeasures are ultimately outlined to safeguard future BFI sensing deployments.

Index Terms—WiFi sensing, beamforming feedback information, adversarial attack, wireless security.

I. INTRODUCTION

BEAMFORMING Feedback Information (BFI) has recently emerged as a practical and effective signal source

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for WiFi-based wireless sensing. Introduced alongside Multi-User Multiple-Input Multiple-Output (MU-MIMO) in the IEEE 802.11ac standard and subsequent iterations [1], BFI can be accessed directly from commodity WiFi hardware without driver or firmware modifications, which makes it significantly easier to deploy than many CSI-based solutions. Although BFI is a compressed representation of Channel State Information (CSI), prior studies have shown that it still preserves sufficient information for a variety of sensing tasks, including respiration monitoring [2], [3], keystroke inference [4], [5], [6], activity recognition [7], [8], and fine-grained analysis [9]. Since these applications are naturally local and are typically expected to operate in homes, offices, meeting rooms, and other indoor WLAN environments, BFI provides a particularly attractive path toward low-cost sensing by reusing existing WiFi infrastructure.

However, this practicality also introduces a timely and underexplored security problem. As BFI sensing moves toward real-world deployments, the consequences of manipulating its inputs become much more serious. Unlike CSI, which is often constrained by hardware and driver access, BFI frames are routinely transmitted in plaintext by commercial WiFi chipsets and typically lack dedicated protection against malicious forgery. This openness allows nearby commodity devices to capture BFI frames and creates an opportunity for adversaries to actively forge and inject deceptive feedback in order to manipulate sensing outcomes. Such attacks are not merely misclassification examples: in real deployments, they may cause a fall detection system to misinterpret an actual fall as benign motion, suppress alerts in vital-sign monitoring, or mislead behavior-aware authentication systems into accepting unauthorized users. For this reason, securing BFI sensing matters now: the technology is practical enough to support emerging real-world applications, while its open feedback mechanism simultaneously exposes a new and underexplored attack surface.

Adversarial attacks in CSI-based sensing have been explored, but applying them to BFI poses challenges due to two main differences: First, CSI attacks manipulate pilots or waveforms, with pilot alterations [10] requiring control over transmitters and earlier methods [11], [12] altering received signals. Unlike CSI, BFI reveals only compressed steering vectors and average gain, lacking subcarrier-level amplitude and phase. This hampers the use of fine-grained perturbations typical in RF interference attacks [13], [14], which depend on creating dynamic multipath reflections to disrupt CSI subcarrier features. Second, attack methods [15] such as signal blocking lose potency against BFI systems that exploit sparsity to reconstruct missing feedback [8].

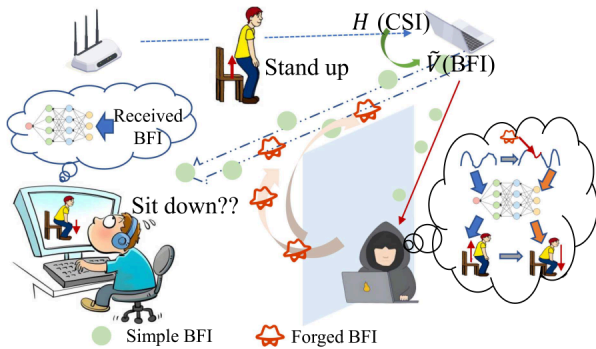


Fig. 1. The STA sends the BFI in plaintext to the AP which can be captured by anyone to perform WiFi sensing. An attacker can inject forged BFI frames to manipulate the BFI sensing system.

Consequently, an entirely new attack paradigm is needed to bridge these gaps.

This paper aims at studying attack methods against BFI sensing to advance its security. As illustrated in Fig. 1, legitimate parties use BFI signals to sense behaviors, while an attacker sniffs these BFI packets and seeks to undermine the sensing outcome. We want to investigate whether an attacker with only off-the-shelf Wi-Fi gear can quietly inject a few BFI frames and trick the sensing system into misreading an action—for example, calling “walking” a “sitting”. To do this, we must solve two simple but tough problems:

- How to make the fake data strong enough to change the result yet still look normal to anyone watching.
- How to cope with the random delays on real networks so the injected packets arrive at exactly the right moment.

MirageBeam’s core insight is to treat the adversarial injection task as a constrained stealth game: craft the minimal set of forged BFI frames that nudge the classifier while remaining statistically indistinguishable from legitimate traffic. This is achieved by segmenting the action into candidate injection points and jointly optimizing the “when” and “what” under a penalty that rewards low frame count and small deviations from natural BFI evolution, ensuring global signal continuity even when the model is deceived. Meanwhile, to cope with the unpredictable latency and timing jitter in real injection process, *MirageBeam* is designed to be robust to injection delays rather than assuming precise synchronization. Instead of targeting a single transmission instant, the attack accounts for a range of possible delays during optimization, allowing the forged beamforming feedback to remain effective even when its injection timing varies. Therefore, the design of *MirageBeam* yields a lightweight attack that preserves both stealth and robustness in practical deployments, without relying on specialized radios or tight timing control.

We implement a working prototype of *MirageBeam* on commodity WiFi devices, demonstrating that the attack can be practically realized without modifying firmware or controlling legitimate transmitters. Furthermore, extensive experiments have demonstrated that *MirageBeam* generalizes across a broad spectrum of threat models and victim systems: it remains effective under both white-box and black-box settings [16], transfers

across heterogeneous learning architectures (like CNN, GRU and Transformer) and classical BFI sensing systems such as BeamSense [7] and MuseFi [8] with distinct feature extraction and inference pipelines.

The main contributions of this paper are as follows:

- We propose *MirageBeam*, the first adversarial framework against BFI sensing, which demonstrates that forged BFI frames can trigger both untargeted and targeted misclassification in commodity WiFi behavior recognition systems.
- We formulate a joint optimization problem that selects forged BFI amplitudes and injection timings to maximize misclassification while enforcing statistical indistinguishability from regular behavior sequences.
- We introduce a delay-aware attack design that maintains effectiveness under real-world injection asynchrony and unpredictable injection latency.
- Extensive evaluations across three indoor environments on off-the-shelf hardware achieve 85.95% (untargeted) and 81.81% (targeted) success rates while maintaining a detection rate below 30%, exposing critical vulnerabilities in current BFI deployments and providing actionable countermeasures.

II. BACKGROUND AND PRELIMINARIES

This section recaps BFI essentials and its role in WiFi behavior recognition.

A. BFI Primer

In a MIMO system with K subcarriers, M_T transmit antennas, and M_R receive antennas, the channel at a given time instant t on the k -th subcarrier is denoted by $\mathbf{H}_k \in \mathbb{C}^{M_R \times M_T}$. The IEEE 802.11ac standard and subsequent iterations [1] introduce a beamforming feedback mechanism. The process begins with the AP (beamformer) transmitting a Null Data Packet (NDP), which contains a known Long Training Field (LTF), together with an NDP Announcement to the client (beamformee). Upon receiving the NDP, the beamformee estimates the channel matrix $\mathbf{H}_k$ for the k -th subcarrier using the known LTF. It then performs Singular Value Decomposition (SVD) on $\mathbf{H}_k$:

$$\mathbf{H}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^H, \quad (1)$$

where $\mathbf{V}_k$ is the right singular matrix. The beamformee extracts the first N_s columns of $\mathbf{V}_k$ to form the beamforming steering matrix $\bar{\mathbf{V}}_k$, where N_s denotes the number of spatial streams. To standardize the feedback format, a phase adjustment is applied so that the last row of each column of $\bar{\mathbf{V}}_k$ becomes real-valued, yielding $\tilde{\mathbf{V}}_k$. Next, $\tilde{\mathbf{V}}_k$ is quantized via Givens Rotation into Beamforming Feedback Angles (BFA), denoted by $\mathbf{a}_k$. These quantized angles are packed into feedback frames and transmitted back to the AP. In our attack setting, the transmitted feedback frames can be captured over the air, and the BFA vector $\mathbf{a}_k$ can then be decoded and reconstructed into $\tilde{\mathbf{V}}_k$. For notational convenience, we refer to such feedback frames as BFI frames, and in the remainder of this paper, BFI refers to the reconstructed phase-adjusted beamforming matrix $\tilde{\mathbf{V}}_k$ unless otherwise stated.

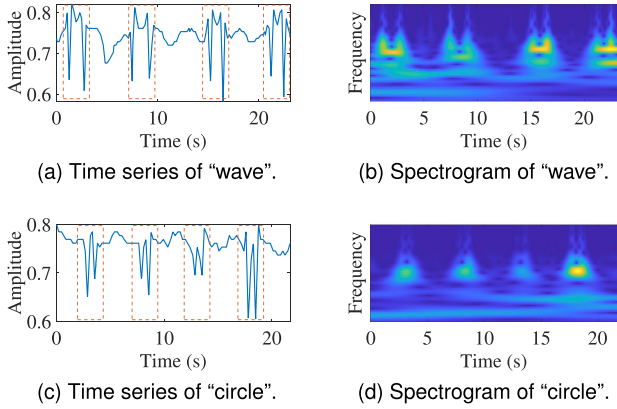


Fig. 2. BFI sequences exhibit distinct regular patterns in both the time domain and frequency domain, depending on the specific behavioral actions.

B. BFI-Based Behavior Recognition

BFI captures motion-induced signal variations (e.g., power attenuation, phase shifts) and forms temporal sequences encoding user motions, represented as 4D tensors $T_N \times M_T \times N_s \times K$, where T_N is frame count. For example, Fig. 2 shows the waveform and time-frequency diagram of the “wave” and “circle” gestures. Modern systems usually adopt deep learning for behavior recognition: BeamSense [7] uses CNN blocks to process BFI sequences directly, while state-of-the-art MuseFi [8] fuses temporal-frequency features with attention for robust classification. Formally, raw input $\mathcal{V}$ undergoes preprocessing $f_{pre}(\cdot)$ (e.g., resampling [5], [8], [17]) to yield standardized input, which is fed to classifier $\mathcal{F}(\cdot)$ with parameters θ . And the final prediction is given by $y = \arg\max \mathcal{F}(x)$.

III. THREAT MODEL AND FEASIBILITY STUDY

This section outlines the threat model and validates the feasibility of hijacking behavior recognition via forged BFI frames.

A. Threat Model

1) *Attack Scenario*: We consider indoor scenarios where an attacker manipulates BFI-based sensing systems by injecting forged BFI frames via a commodity WiFi NIC. In this work, we focus on the common single-person WiFi sensing setting, where at most one active user performs the target activity. As shown in Fig. 3, the attacker first passively collects BFI data from the target environment, such as public datasets or local overhearing, then uses a lightweight optimization algorithm to craft stealthy perturbations that alter the system’s sensing decisions, and finally encodes the forged BFI into the standard feedback format for injection at selected time points. In our considered defense setting, the defender examines suspicious BFI inputs before sensing inference, mainly through content-level detection, timing-level detection, protocol-level validation, and anomaly-based sample identification, which will be discussed in Sections VII and VIII-E.

2) *Attacker’s Capabilities*: The attacker has two core capabilities. First, the attacker is assumed to obtain coarse action-start synchronization. *MirageBeam* does not require

tight time synchronization. An attacker can passively monitor over-the-air packets with commodity NICs and trigger injection via lightweight onset detection on the BFI stream. In indoor WLANs, timing uncertainty is mainly caused by MAC contention/backoff and queuing, typically at the tens-of-milliseconds level with occasional long tails under heavy load [18]. Our method explicitly models the onset-to-injection offset as a window-level shift and optimizes the injection schedule accordingly, tolerating moderate delays without materially degrading attack success. Second, the attacker is equipped with two-dimensional attack preparation capabilities, which consist of model access and label availability. These two capabilities are elaborated as follows:

a) *Model Access Capability*: We consider both white-box and black-box settings, covering the full range of practical attacker model access possibilities. In the white-box setting, the attacker obtains complete access to the victim sensing model $\mathcal{F}(\cdot)$ including architecture and parameters. This is reasonable since many WiFi sensing models, such as WiDar [19], are publicly available and widely reused. In the black-box setting, the attacker cannot directly access $\mathcal{F}(\cdot)$; instead, they train a local surrogate model and generate transferable adversarial samples by leveraging the transferability of learned representations [16]. This is a common and valid approach for black-box attack scenarios.

b) *Label Availability*: We do not assume the attacker can directly obtain accurate semantic labels, but hold that partial label information can be inferred via observation and clustering methods, which conforms to practical attack scenarios. The attacker passively collects unlabeled BFI samples and trains an auxiliary unsupervised model $\mathcal{G}(\cdot)$ to cluster samples in the latent feature space. They can either infer semantic labels by associating clustered samples with real-world activities through system response observation, or adopt cluster identifiers as surrogate labels when semantic information is unavailable. Such labels are sufficient for defining attack objectives. For instance, in intrusion detection, the attacker only needs to mislead intrusion-related samples into the “no-intrusion” cluster, which can be easily identified by quasi-static and low-variation signal characteristics and aligned with a specific cluster of $\mathcal{G}(\cdot)$. Relevant validation is presented in Section VIII-A.

The full white-box access is not required for *MirageBeam*. We explicitly consider two representative settings. *White-box* serves as an upper bound: the attacker knows the victim model $\mathcal{F}(\cdot)$ (architecture and parameters) and has access to the relevant labels, enabling direct optimization to generate forged BFI sequences. *Black-box* is more practical: the attacker cannot access $\mathcal{F}(\cdot)$ nor the victim’s labeled dataset; instead, they train a local surrogate model and exploit transferability (Section VI-F). The supervision needed to drive optimization or train the surrogate can be obtained from (i) true labels inferred from observable user activities, or (ii) pseudo-labels derived via clustering/representation learning (Section VIII-A), under which effective attacks remain feasible.

3) *Attack Types*: Let $f_{pre}(\cdot)$ denote the preprocessing function for the BFI-based sensing system. For an original BFI

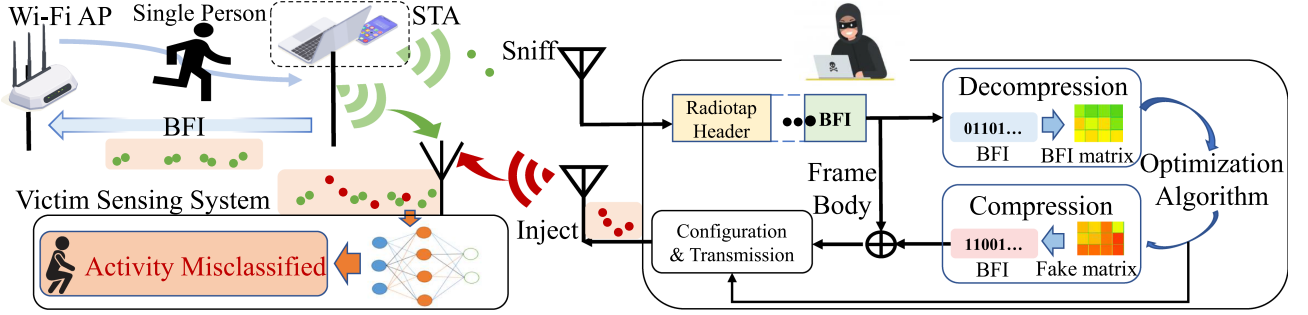


Fig. 3. Attacker scenario: the attacker injects forged BFI frames to manipulate sensing results.

sequence $\mathcal{V}$, the preprocessed input is given by

$$x = f_{\text{pre}}(\mathcal{V}), \quad (2)$$

and let $\mathcal{F}(\cdot)$ represent the target sensing classifier that performs decision-making on the preprocessed input x . When an attacker injects a forged signal s via the injection operation $\mathcal{I}(\cdot)$, the forged sequence is further preprocessed to generate an adversarial sample as

$$x' = f_{\text{pre}}(\mathcal{I}(\mathcal{V}, s)). \quad (3)$$

All attack objectives are defined based on the decision behavior of the target classifier $\mathcal{F}(\cdot)$ on the original input x and adversarial input x' . We divide the adversarial attacks into two typical types according to distinct attack purposes:

a) *Untargeted Attack*: The goal of an untargeted attack is to disrupt the decision consistency of $\mathcal{F}(\cdot)$. For an original input x , the attacker constructs the adversarial input x' such that

$$\mathcal{F}(x') \neq \mathcal{F}(x), \quad (4)$$

where the decision output of $\mathcal{F}(\cdot)$ can correspond to either a semantic activity label or a cluster-based surrogate label.

b) *Targeted Attack*: The goal of a targeted attack is to steer the output of $\mathcal{F}(\cdot)$ from its original decision to a specific desired decision region. Formally, the attacker constructs x' to satisfy

$$\mathcal{F}(x') = r_t, \quad (5)$$

where r_t denotes the desired decision outcome of $\mathcal{F}(\cdot)$, and r_t can be either a semantically inferred activity label or a cluster-derived surrogate label from $\mathcal{G}(\cdot)$.

B. Feasibility Study

We study the attack feasibility from two perspectives: the forgeability and stealth of BFI.

Feasibility of Forged BFI Injection: The feasibility of adversarial BFI injection hinges on the ability to accurately reconstruct and inject forged BFI frames. This process is validated through a systematic implementation and verification workflow: (1) *End-to-end Pipeline*: Using commodity hardware, we compress a designed BFI matrix into valid bytes, embed them into legitimate frames, and inject over the air. (2) *Preliminary Test*: In a 4×1 MU-MIMO setup, two forged frames were replayed with a 0.7 s interval. (3) *Results* (Fig. 4): Received frames maintained the interval, aligned closely with injected ones, and

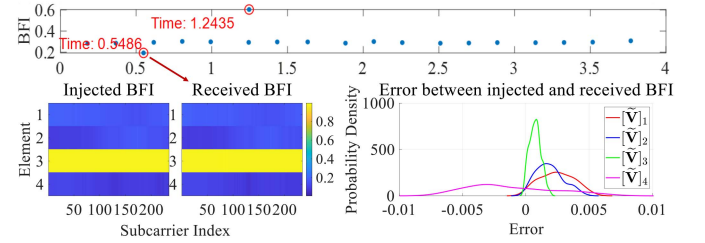


Fig. 4. Feasibility demonstration of forged BFI injection: BFI can be successfully replayed with a small margin of error.

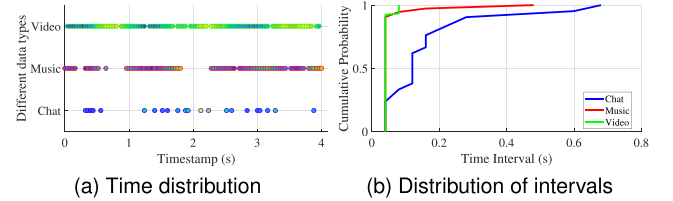


Fig. 5. Concealment demonstration of BFI injection: the rate of BFI is unstable, showing sparseness in time and uncertainty in time intervals.

showed minimal element-wise errors (attributed to compression quantization, not affecting integrity). This confirms high-fidelity forgery, injection, and reception of BFI frames.

Concealment of BFI Injection: BFI injection is inherently covert. The 802.11 protocol does not mandate a fixed interval for BFI reporting, and BFI transmissions are often event-triggered. Additionally, real-world background traffic introduces natural irregularity in BFI timing. Fig. 5 presents BFI arrival patterns under three common usage scenarios: video streaming, music playback, and text chatting [5]. High-throughput activities like video streaming lead to dense and continuous BFI sequences, while lighter traffic scenarios produce sparse and irregular sequences. In all cases, the temporal distribution of BFI is non-uniform and bursty, allowing forged frames to blend seamlessly into legitimate traffic patterns without raising suspicion.

These preliminary experiments confirm that BFI frames can be forged and injected, and that unstable packet rates also allow them to be ignored. Yet a successful attack demands a large injection volume and conspicuous deviations, which sharply raise the risk of detection. Thus, the critical task is to craft subtle BFI frame forgery and injection strategies that remain effective while minimizing exposure.

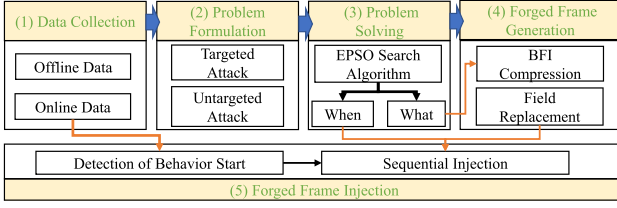
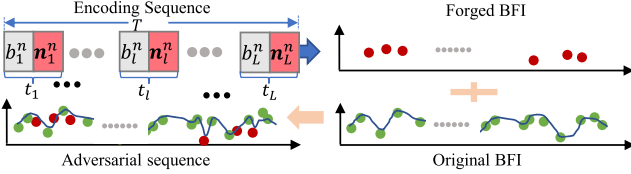
Fig. 6. Overview of the *MirageBeam* design.

Fig. 7. The process of generating attack signals.

IV. ATTACK DESIGN OF MIRAGEBEAM

This section details *MirageBeam*'s five-stage design with a two-stage pipeline. As shown in Fig. 6, offline it collects legitimate BFI samples per target behavior, uses Enhanced Particle Swarm Optimization (EPSO) to solve a constrained optimization problem for stealthy forged BFI generation which is encoded into standard-compliant frames; online, a lightweight trigger detects user activity onset to inject precomputed frames, altering sensing system observations and misleading predictions.

A. Data Collection

MirageBeam uses two BFI sets: offline templates per behavior class for crafting fakes, and live streams with motion triggers for timely injection.

1) *Offline Data*: To enable targeted adversarial injection, *MirageBeam* first collects legitimate BFI sequences for each behavior class (e.g., $\mathcal{V}^y$ for class y) via passive monitoring. This offline data is then used to generate forged and adversarial BFI sequences matching the legitimate signal distribution.

2) *Online Data*: During deployment, *MirageBeam* monitors real-time BFI streams and uses motion-trigger mechanisms to detect the start of user activity by identifying sudden BFI signal changes or patterns. This behavior onset detection ensures timing for injecting pre-prepared forged BFI frames.

B. Problem Formulation

Forged BFI frames are generated by encoding the reference BFI matrix. To generate optimal forged frames, we first need to establish an optimization problem. The problem modeling needs to clarify two aspects: one is the optimization variables and their encoding methods, and the other is the optimization objectives and constraints.

Optimization variables and their encoding methods: To mathematically model the two core optimization variables in the attack—*when to inject* and *what to inject*—we design an encoding mechanism (Fig. 7) as follows:

- *When to inject*: Each behavior's time window (duration T) is uniformly divided into L segments, with midpoints t_1 to

t_L serving as candidate injection timings. In the paper, we set L to 50.

- *What to inject*: For each candidate time t_l , we optimize a pair $\{b_l, \mathbf{n}_l\}$: $b_l \in \{0, 1\}$: A binary flag determining whether to inject a forged frame at t_l ; $\mathbf{n}_l \in \mathbb{C}^{M_T \times 1}$: A complex vector specifying the core perturbation component for constructing the forged frame.

From $\mathbf{n}_l$ to forged BFI: Given a reference BFI matrix $\mathbf{d}_{\text{base}}$ collected from legitimate samples, the forged BFI $\mathbf{d}_l \in \mathbb{C}^{M_T \times K}$ to be injected at time t_l is constructed by

$$\mathbf{d}_l = \text{IFFT}([\mathbf{n}_l, \text{FFT}(\mathbf{d}_{\text{base}})_{:,2:K}]). \quad (6)$$

For clarity, the FFT and IFFT are both applied along the subcarrier axis K , independently for each antenna row of $\mathbf{d}_{\text{base}} \in \mathbb{C}^{M_T \times K}$. The variable $\mathbf{n}_l \in \mathbb{C}^{M_T \times 1}$ contains one complex coefficient for each antenna row, and $[\mathbf{n}_l, \text{FFT}(\mathbf{d}_{\text{base}})_{:,2:K}]$ denotes column-wise concatenation along the subcarrier dimension, i.e., replacing the first column of the transformed matrix while keeping the remaining columns unchanged. This transform-domain operation is introduced as a compact structured parametrization to generate sufficient variation in the forged BFI, rather than to imply a specific physical interpretation. The IFFT is then applied along the same subcarrier axis for each antenna row to reconstruct the forged matrix. Since $\tilde{\mathbf{V}}_k$ consists of the first N_s columns of the unitary matrix, the power sum of each column in $\mathbf{d}_l$ should be 1, i.e., $\sum_{m=1}^{M_T} |\mathbf{d}_l(m, k)|^2 = 1$. Thus, the generated forged BFI should also be normalized column by column. This encoding method is motivated by the considerations: We employ a low-frequency transform-domain parameterization to balance attack efficacy and computational efficiency. For each injected timing, a forged BFI is a tensor of size $M_T \times K \times N_s$ (Tx antennas $\times$ subcarriers $\times$ spatial streams). Optimizing it directly requires $M_T K N_s$ variables per timing. By constraining the optimization to low-frequency coefficients along the subcarrier dimension, we retain the ability to induce coherent, structured changes to the instantaneous BFI pattern (sufficient to reshape the reconstructed sensing waveform and cause misclassification), while reducing the variables to $M_T N_s$ per timing. With L_R injected timings, the total variable count decreases from $L_R M_T K N_s$ to $L_R M_T N_s$, i.e., an $\mathcal{O}(K)$ reduction, which substantially improves convergence speed and practical feasibility. Therefore, the real-world forged signal is composed based on the forged BFI sequence $\mathbf{d} = [\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_L]$ and the selection sequence $\mathbf{b} = [b_1, b_2, \dots, b_L]$.

Optimization objectives and constraints: To evaluate each candidate injection signal, we use the classifier's logit scores, which are the raw outputs before softmax. Compared to probabilities, logit score provides more stable and informative feedback for optimization, which is defined as:

$$F(x)_i = \text{Fit}^i(\mathcal{F}(f_{\text{pre}}(\mathcal{V}))) = \text{fit}^i(\mathcal{V}), \quad (7)$$

where $\text{Fit}^i(\mathcal{F}(\cdot))$ represents the confidence that the input $x = f_{\text{pre}}(\mathcal{V})$ is classified as the label i under the classifier $\mathcal{F}(\cdot)$.

We further formulate optimization problems for targeted and untargeted attacks based on the encoded injection signal s and logit scores.

1) *Untargeted Attack*: The goal is to induce misclassification of the input $\mathcal{V}^y$ with ground truth label y , by injecting a sparse set of forged BFI frames via an injection strategy $s^{y \rightarrow \bar{y}}$. Let $x' = f_{pre}(\mathcal{I}(\mathcal{V}^y, s^{y \rightarrow \bar{y}}))$ denote the adversarial sample. The optimization problem is defined as:

$$\underset{s^{y \rightarrow \bar{y}}}{\text{minimize}} \quad c_1 \cdot \mathcal{L}_u(x') + c_2 \cdot \|\mathbf{b}\|_1 + c_3 \cdot \|\delta\|_F, \quad (8a)$$

$$\text{s.t.} \quad C1: b_i \in \{0, 1\}, \quad C2: \sum_{m=1}^{M_T} |\mathbf{d}_i(m, k)|^2 = 1, \quad (8b)$$

where $\mathcal{L}_u(x')$ encourages misclassification by maximizing the logit margin between any incorrect class and the true label and $\|\mathbf{b}\|_1$ counts the number of injected BFI frames (sparsity constraint). c_1 , c_2 and c_3 are weighting hyperparameters. $\|\delta\|_F$ measures the cumulative distortion introduced by forged signals, which is defined as:

$$\|\delta\|_F = \sum_{i=1}^{\|\mathbf{b}\|_1} (\|\mathbf{d}_i - \mathbf{V}_{le(i)}\|_F + \|\mathbf{d}_i - \mathbf{V}_{ri(i)}\|_F), \quad (9)$$

where $\mathbf{d}_i$ is the i -th injected BFI frame, and $\mathbf{V}_{le(i)}$, $\mathbf{V}_{ri(i)}$ are the temporally adjacent original frames to the left and right of $\mathbf{d}_i$, respectively. This formulation encourages the injected frames to remain consistent with the local temporal structure of legitimate traffic. The *Fitness Score* of the current individual $s^{y \rightarrow \bar{y}}$ is calculated by (8a). The untargeted loss [20] is defined as:

$$\mathcal{L}_u(x') = \min(\max_{i \neq y} \{F(x')_i\} - F(x')_y, \xi), \quad (10)$$

where ξ is a constant that controls the misclassification confidence.

2) *Targeted Attack*: In targeted attacks, the adversary aims to force a sample $\mathcal{V}^y$ to be classified as a specific target label $y^t \neq y$. The corresponding optimization problem becomes:

$$\underset{s^{y \rightarrow y^t}}{\text{minimize}} \quad c_1 \cdot \mathcal{L}_t(x') + c_2 \cdot \|\mathbf{b}\|_1 + c_3 \cdot \|\delta\|_F, \quad (11a)$$

$$\text{s.t.} \quad C1, C2, \quad (11b)$$

where $x' = f_{pre}(\mathcal{I}(\mathcal{V}^y, s^{y \rightarrow y^t}))$ is the adversarial input. The targeted loss function is:

$$\mathcal{L}_t(x') = \min(F(x')_{y^t} - \max_{i \neq y^t} \{F(x')_i\}, \xi). \quad (12)$$

This loss encourages the logit of the target class y^t to exceed all others by a margin, capped by ξ for numerical stability. The *Fitness Score* of the current individual $s^{y \rightarrow y^t}$ is calculated by (11a).

Actually, the candidate injection timings are optimized *jointly* within each behavior window. This joint formulation allows the optimizer to balance attack effectiveness and injection sparsity under a single objective while keeping the problem size bounded by L per window. It is worth noting that this joint optimization is performed in the offline attack-generation stage. Therefore, the joint treatment of candidate timings increases offline optimization cost, but does not alter the lightweight nature of the online attack stage.

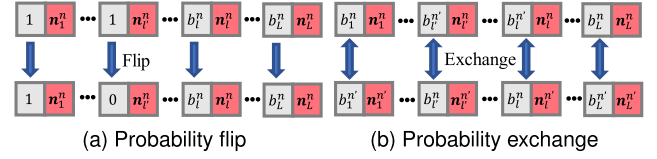


Fig. 8. The proposed crossover-mutation operations in EPSO.

C. Problem Solving

Since the insertion instants of forged BFI frames are discrete, the objectives in (8) and (11) are non-differentiable with respect to this variable, rendering conventional gradient-based optimizers ineffective. We therefore solve these problems with an Enhanced Particle Swarm Optimization (EPSO) tailored for mixed continuous-discrete variables, whose gradient-free global search quickly handles hybrid decision spaces. The algorithm's solution procedure proceeds as follows.

Initialization: The attacker begins by randomly generating an initial population comprising N individuals.

Fitness calculation: The attacker calculates the fitness score of every injection signal s_i in the generation.

Updating: The speed and position (strategy) of each particle are updated according to the following formula:

$$v_i^{(t+1)} = w \cdot v_i^{(t)} + c_1 \cdot r_1 \cdot (sp_i - x_i^{(t)}) + c_2 \cdot r_2 \cdot (sg - x_i^{(t)}) \quad (13)$$

and

$$s_i^{(t+1)} = s_i^{(t)} + v_i^{(t+1)}, \quad (14)$$

where $v_i^{(t)}$ is the velocity of the i -th particle at iteration t , w is the inertia weight, c_1 and c_2 are the individual and social learning factors, r_1 and r_2 are random numbers in $[0, 1]$, sp_i is the personal best position of the i -th particle, sg is the global best position of the swarm, and $x_i^{(t)}$ is the position of the i -th particle at iteration t . The inertia weight w balances exploration versus exploitation, and the cognitive and social coefficients c_1 and c_2 determine the relative influence of personal and global best positions. Prior studies [21], [22] recommend $w \in [0.5, 1]$ and $c_1 = c_2 \approx 1.5$, which justify our settings ($c_1 = c_2 = 1.5, w = 0.5$) in *MirageBeam*.

Crossover-Mutation: The presence of discrete variables $\mathbf{b}$ limits the search efficiency of traditional PSO. To address this, we introduce a crossover-mutation mechanism that enhances PSO's exploration while adhering to variable constraints. Each individual undergoes crossover-mutation with a probability of 0.2. As illustrated in Fig. 8, we employ two operations: (1) *Probability flipping*: Each bit in $\mathbf{b}$ is flipped ($0 \leftrightarrow 1$) with a probability of 0.2. (2) *Probability exchange*: With a probability of 0.2, two individuals are selected, and their $\mathbf{b}$ elements are swapped at corresponding positions. We set the probabilities of these operations to 0.2, balancing exploration and exploitation, whose choice is consistent with prior works [23], [24] that adopt similar probabilities in evolutionary optimization and demonstrate effectiveness in practice.

In this algorithm, initialization runs once; iterations update the swarm and re-score each member. The fittest individual's

decoded sequence yields the solved optimal timing and contents of the forged BFI. Crafting one adversarial sample/window is performed by a population-based optimization procedure. The average runtime per optimization iteration is 12.5 s, and the average convergence point is 131 iterations, resulting in an average total crafting time of $12.5 \times 131 = 1637.5$ s (≈ 27.3 min) per sample/window. All experiments were conducted at the Big Data Computing Center of Southeast University on a CPU node (Intel Xeon 6126, 24 cores, 192 GB RAM). Given the current computational demands, the optimization process in *MirageBeam* can only be performed offline in its current form. Optimization is performed offline, while the online phase only uses pre-computed parameters to perform lightweight start-up triggering, timing alignment, and frame injection.

D. Forged Frame Generation and Injection

The forged BFI content is first compressed via Givens Rotation and decomposed into angle values $\mathbf{a}_k$. These angles are converted to binary, with bit-order reversals applied at both the bit and 8-bit segment levels, then transformed into hexadecimal bytes. These bytes replace the BFI fields in legitimately captured over-the-air frames, generating forged BFI frames that are then injected upon detecting user activity onset. The online injection consists of two components:

1) *Activity Onset Detection*: *MirageBeam*'s trigger continuously monitors real-time BFI streams to detect user activity onset. It identifies activity starts by tracking BFI pattern anomalies such as sudden amplitude or phase shifts that mark the target behavior's beginning, as common in existing WiFi sensing [5], [25]. The current prototype does not rely on a separately implemented onset-detection module with explicitly benchmarked thresholds. Instead, in each trial, the beginning of the human action, signal capture, and forged-frame injection are synchronously initiated by the experimental procedure. This setting provides coarse synchronization around the action-start region.

2) *Sequential Injection*: In our implementation, the attacker first filters over-the-air BFI frames by the target link's MAC address/BSSID using standard packet-capture tools, so that BFI traffic from other APs or clients can be excluded before forging. The forged BFI frames are then synthesized based on the same target MAC context and injected toward the sensing receiver. Upon detecting activity onset, *MirageBeam* injects a sequence of precomputed forged BFI frames according to an optimized timing schedule determined during the offline phase. These forged frames are injected incrementally and captured alongside natural BFI frames by the receiver, forming an adversarial BFI sequence that misleads the sensing system toward incorrect predictions. During online deployment, only trigger-time detection and forged-frame injection are performed, and the latency from trigger to the first injected frame is approximately 20 ms in our prototype.

V. ROBUSTNESS ENHANCEMENT

This section addresses challenges in real-world injection, proposing enhancements to boost *MirageBeam*'s robustness.

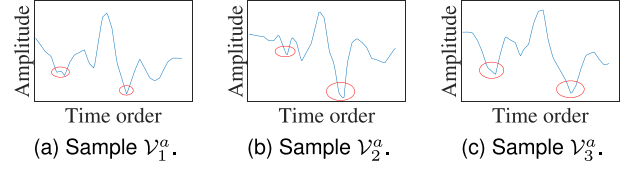


Fig. 9. Three normal BFI samples of 'walk' with differences in red circles.

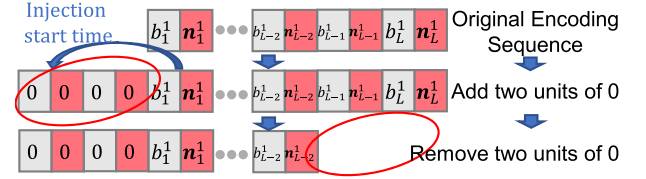


Fig. 10. Delay window with $T_d = 2$.

A. Challenges in Real-World Injection

In real-world deployment, unpredictable delays cause injection asynchrony and signal variability reduces attack success and generalizability.

1) *Injection Delay*: Uncontrollable delays disrupt temporal synchronization between forged signals and target behaviors. These delays derive from: *Hardware latency*: Processing delays in the attacker's NIC and receiver-side signal handling; *Propagation latency*: Signal travel time, influenced by distance and obstacles. Such delays cause forged frames to arrive after the intended injection time, decoupling them from the natural behavior sequence and reducing attack efficacy.

2) *Signal Variability*: BFI patterns are prone to distortions that invalidate precomputed forgeries as shown in Fig. 9, driven by: *Environmental noise*: Multipath interference and dynamic obstacles introducing random signal fluctuations; *User heterogeneity*: Individual differences in movement (e.g., gait, speed) causing BFI change even for identical actions. These adverse factors break generalization: forgeries optimized for one BFI sample $\mathcal{V}_1^y$ often fail for others ($\mathcal{V}_2^y$) of the same behavior.

B. Enhancement Measures

1) *Delayed-BFI Injection Window Mechanism*: In order to suppress the detrimental effects caused by unstable delays, we try to introduce a delayed-BFI injection window mechanism during the optimization process to improve the tolerance of the optimization results to the delay. As illustrated in Fig. 10, we prepend T_d units of 0 to the beginning of the interference signal sequence, resulting in a new sequence length of $L + T_d$. Subsequently, T_d units are removed from the tail of the new sequence, yielding an interference signal sequence with a delay of T_d units. This modified signal corresponding to the delay window of length T_d simulates the actual delay. If we define the process of delaying the interference signal s by p units as $De(s, p)$, the adversarial sample can be expressed as

$$x' = f_{pre}(\mathcal{I}(\mathcal{V}, De(s, p))), \quad (15)$$

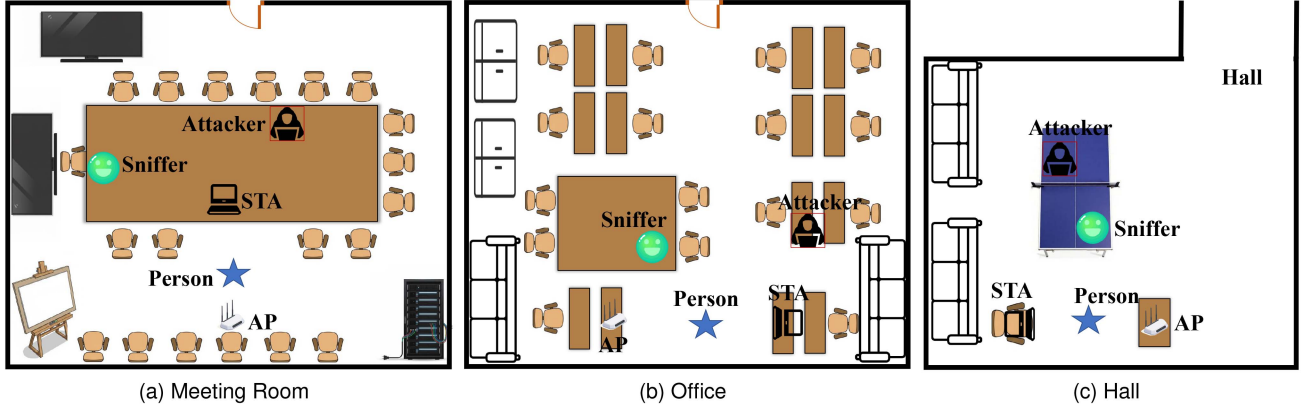


Fig. 11. Experiment setup in three environments.

and the “logit score” in (7) can be updated as follows:

$$F_{en}(x')_i = \sum_{p=0}^{T_d} w_p \cdot fit^i(\mathcal{I}(\mathcal{V}_j^y, De(s, p))), \quad (16)$$

where $w_p \in [0, 1]$ denotes the weight for the p -unit delayed sequence and $\sum_{p=0}^{T_d} w_p = 1$.

2) *Multi-Sample Adaptation*: To address the challenge of sample diversity, we further enhance the objective function. Specifically, an attacker can take a batch of samples under different distortion environments and apply disturbance s to each sample, and finally sum the score of all disturbed samples to improve the robustness of the disturbance signal. The “enhanced logit score” in (16) can be formulated as

$$F_{en}(x')_i = \sum_{j=1}^{N_b} \sum_{p=0}^{T_d} w_p \cdot fit^i(\mathcal{I}(\mathcal{V}_j^y, De(s, p))), \quad (17)$$

where N_b is the number of the batch. In the paper, we set it to 20.

C. Final Optimization Problem

Therefore, the final optimization problem for *untargeted attacks* can be reformulated as:

$$\underset{s^y \rightarrow \bar{y}}{\text{minimize}} \quad c_1 \cdot \mathcal{L}_u(x') + c_2 \cdot \|\mathbf{b}\|_1 + c_3 \cdot \|\delta\|_F, \quad (18a)$$

$$\text{s.t. } C1, C2, C3 : w_p \in [0, 1] \text{ and } \sum_{p=0}^{T_d} w_p = 1 \quad (18b)$$

where

$$\mathcal{L}_u(x') = \min(\max_{i \neq y} \{F_{en}(x')_i\} - F_{en}(x')_y, \xi). \quad (19)$$

And the *targeted attack* loss function is shown in the following equation:

$$\mathcal{L}_t(x') = \min(F_{en}(x')_{y^t} - \max_{i \neq y^t} \{F_{en}(x')_i\}, \xi). \quad (20)$$

This final optimization problem integrates key objectives of misclassification induction and stealth, while accounting for real-world constraints.

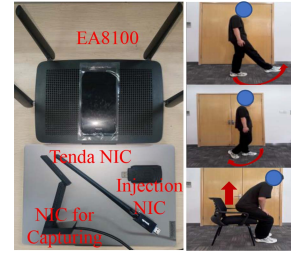


Fig. 12. Devices and Motion.

VI. EVALUATION

This section first outlines the experiment setup, then presents experimental results from four aspects: attack performance, comparison with random injection, performance on existing systems, and black-box attack.

A. Experiment Setup

Hardware Setup: We deployed a Linksys EA8100 as AP, a laptop with Tenda U10 (IEEE 802.11ac) as STA, operating on channel 153 (5.765 GHz, 80 MHz). A Netgear A6210 acted as a sniffer for legitimate BFI capture, while an Ubuntu 20.04 computer with a monitor-mode dual-band NIC served as an attacker to capture normal BFI and inject forged frames, using Scapy for frame parsing, modification, and replay [26]. The devices and motion are shown in Fig. 12.

Data Collection: As shown in Fig. 11, *MirageBeam* was tested in three environments: meeting room, office, hall. We collected data on three behaviors: walking (W), walking backward (WB), standing up (S). Forged BFI generation and injection were implemented in Python; we broadcast forged frames captured by receivers during activities. In total, 2023 normal BFI samples (for training) and 3119 adversarial samples were gathered, each 2.8 s long. The data collection process lasted a month and a half. These experiments have strictly followed our university’s Institutional Review Board (IRB).

Learning Model: We use a compact CNN for three-class classification: two convolutional layers (32, 64 filters) with max-pooling, a 128-unit fully connected layer with dropout, and an output layer for three classes. Trained with Adam and

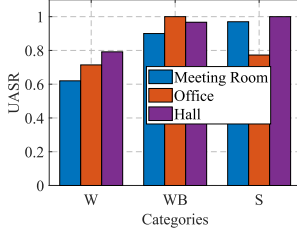


Fig. 13. Overall UASR.

categorical cross-entropy loss. We also test advanced BFI systems BeamSense [7] and MuseFi [8].

Evaluation metrics: Untargeted success rate (UASR) is the probability of misclassification: $\text{UASR} = \frac{\sum_{i=1}^{N_{\text{total}}} \zeta(y_i \neq y)}{N_{\text{total}}}$, where $\zeta(\cdot)$ is 1 for success. Targeted success rate (TASR) is $\text{TASR} = \frac{\sum_{i=1}^{N_{\text{total}}} \zeta(y_i = y^t)}{N_{\text{total}}}$ for specific labels.

B. Performance of Attack

1) **Overall Performance:** Fig. 14(a) shows the TASR across three environments. *MirageBeam* consistently achieves over 75% accuracy in inducing the classifier to predict the attacker-specified label. The confusion matrices (Fig. 14(b) to (d)) reveal that the injected adversarial BFI sequences effectively mislead the classifier, with minimal confusion among unintended classes. Additionally, Fig. 13 reports the UASR, where any incorrect classification is considered successful. *MirageBeam*'s TASR in meeting room, office, and hall scenarios are 83.04%, 75.46%, and 86.94%, respectively, and its UASR are 83%, 82.9%, and 91.96%, respectively.

2) **Stealthiness Analysis:** To evaluate the visual impact of our stealthiness-aware adversarial signal design, we compare the perturbed BFI sequences generated with and without the concealment constraint. As shown in Fig. 15(a) to (c), the 3D visualizations illustrate the original BFI sequence, the perturbed sequence generated without concealment, and the one generated by our method. It is evident that our method introduces significantly less distortion in both time and subcarrier dimensions, closely resembling the original BFI structure. Fig. 16 illustrates the comparison between a design without stealthiness and *MirageBeam*. It is evident that both the MSE and the total number of injected frames for *MirageBeam* are significantly lower than in the scenario lacking a stealthiness design; however, the attack accuracy is slightly lower. We further quantified this stealthiness metric in Section VII-A.

3) **Robustness Analysis:** To further evaluate the robustness of *MirageBeam*, we conduct comparative experiments with and without robustness design. Specifically, we examine whether incorporating robustness mechanisms such as delay modeling and multi-sample adaptation can enhance attack performance across varying environments. As shown in Fig. 17(a), *MirageBeam* achieves higher TASR with similar MSE across all action categories compared to the baseline without robustness design. A similar trend is observed in Fig. 17(b) for the UASR, where our method consistently yields improved misclassification effectiveness.

TABLE I
COMPARISON OF BLACK-BOX ATTACK PERFORMANCE

Model	Before Enhancement		After Enhancement	
	UASR	TASR	UASR	TASR
5-layer CNN	78.41%	75.17%	88.30%	85.75%
GRU	16.83%	10.80%	65.09%	49.06%
Transformer	67.14%	59.09%	88.94%	83.73%

4) **Non-Line-of-Sight Attack:** To evaluate *MirageBeam* under more realistic conditions, we analyzed its both targeted (TA) and untargeted (UA) attack success rates in both line-of-sight (LoS) and non-line-of-sight (NLoS) scenarios. In the NLoS setting, the direct path between the attacker and the legitimate receiver was obstructed using cardboard. Fig. 18 show that while the success rate in NLoS is slightly reduced compared to LoS, the overall effectiveness remains comparable. This robustness arises from *MirageBeam*'s use of frame injection rather than manipulation of the original BFI frame elements. As long as the injected signal at the receiver exceeds the reception threshold, the attack can succeed. These findings indicate that *MirageBeam* is resilient even in obstructed environments, highlighting its practicality in real-world scenarios.

5) **Impact of Devices:** To further examine hardware-related generalization, we extended the experiment beyond the original setup by introducing another AP device and a different STA-side wireless adapter. Specifically, in addition to the original AP₁–STA₁ setting, we added AP₁–STA₂ and AP₂–STA₁, where AP₂ is a Linksys EA8100 and STA₂ uses a TP-Link adapter, different from the original Tenda U10 used in STA₁. As shown in Fig. 19, the proposed attack remains consistently effective across these hardware combinations. These results provide stronger evidence that the attack is not tied to a single AP or a single client adapter.

6) **Impact of Distance:** As shown in Fig. 20(b), we evaluated the attack success rates at varying distances between the attacker and the legitimate receiver, specifically at 1, 5, and 10 meters. The results indicate that both targeted and untargeted attacks consistently achieve success rates above 80%, with minimal variation across distances. This demonstrates that as long as the injected frame can be successfully captured by the receiver, the attack remains effective, which shows the excellent adaptability of *MirageBeam* at different distances.

7) **Impact of Person Variety:** User diversity in terms of body shape and behavioral habits may affect the effectiveness of *MirageBeam*. To further examine person-specific bias, we expanded this experiment to five volunteers under the same environmental conditions. As shown in Fig. 20(b), both targeted and untargeted attacks remain consistently effective across all five victim identities. The attack accuracy stays around or above 70% for all participants, and reaches above 80% for most cases, which shows that the proposed method is not restricted to the original three subjects. Overall, these results provide stronger support that *MirageBeam* can generalize across different individuals and alleviate the concern of person-specific bias.

8) **Impact of Channel:** To further examine channel-related generalization, we additionally evaluated the proposed attack under two different channel settings, namely channel 40 and

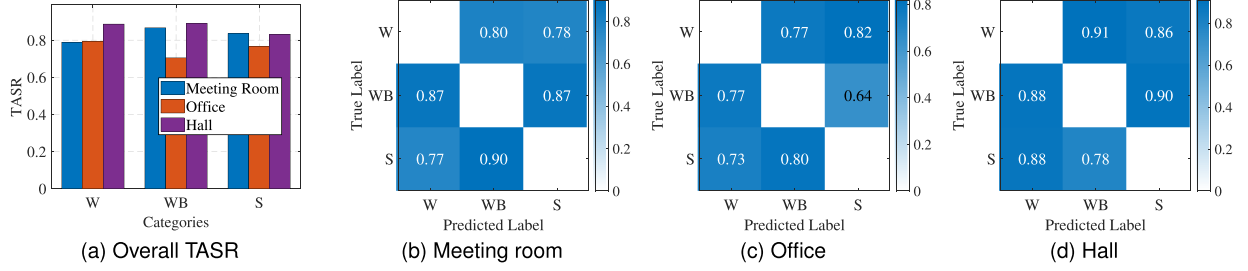


Fig. 14. The overall targeted attack success rate for activity recognition.

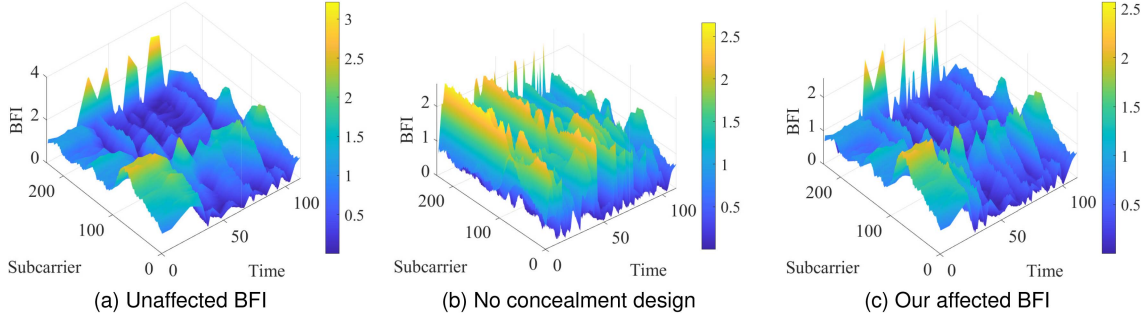


Fig. 15. 3D images and MSE distribution between the affected samples and the original samples.

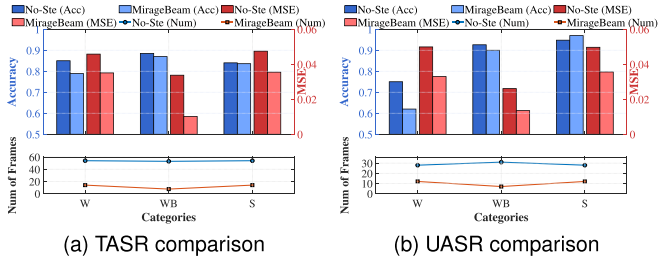


Fig. 16. Stealthiness comparison results.

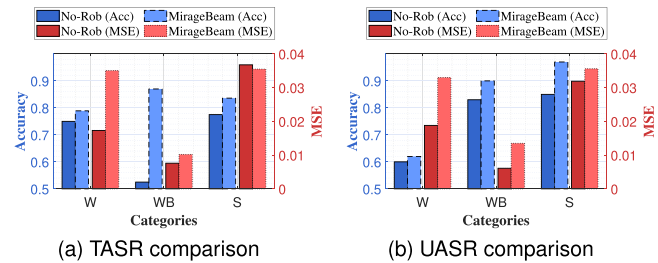


Fig. 17. Robustness comparison results.

channel 153. As shown in Fig. 20(c), both targeted and untargeted attacks remain consistently effective across the two channels, with attack success rates staying at a relatively high level. These results provide further evidence that the proposed method is not limited to a single channel configuration.

C. Comparison With Random Injection

To validate the effectiveness of our adversarial signal optimization, we compare *MirageBeam* against a random injection

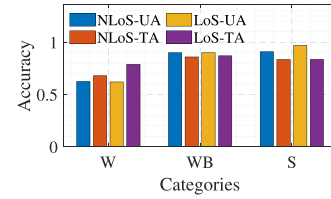


Fig. 18. Attack success rate under LoS and NLoS.

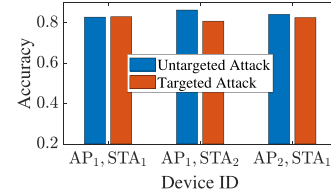


Fig. 19. Attack success rate under different devices.

baseline with the same number of injected BFI samples. Fig. 21 presents both targeted and untargeted attack success rates. The results clearly show that *MirageBeam* significantly outperforms random injection in all cases. For untargeted attacks, *MirageBeam* achieves success rates of 62%, 90%, and 97% for W, WB, and S, respectively, compared to only 2.5%, 57.5%, and 57.5% for the random baseline. For targeted attacks, *MirageBeam* attains 75%, 87%, and 82% accuracy, while random injection yields notably lower success rates (1.25%, 27.5%, and 21.25%). These results highlight the importance of optimization-based adversarial signal generation, as naive random injection is largely ineffective even when matched in injection volume.

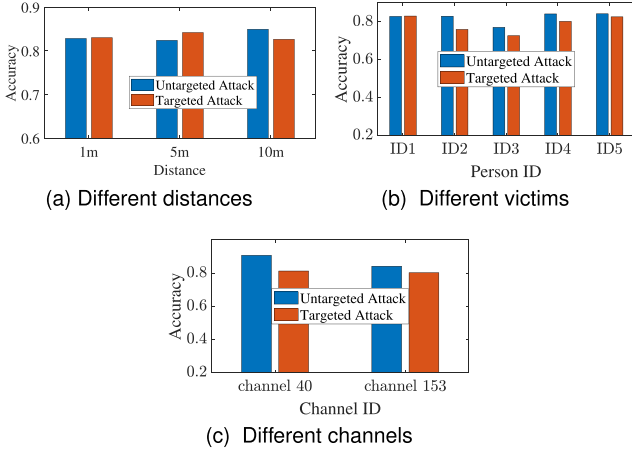
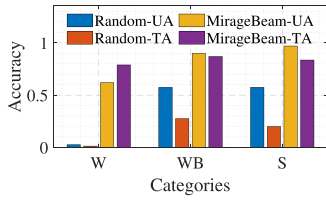
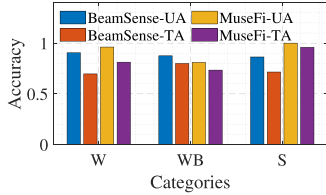


Fig. 20. Attack success rate under different distances, victims and channels.

Fig. 21. *MirageBeam* versus Random Injection.Fig. 22. *MirageBeam* performance on existing systems.

D. Performance on Existing Systems

To evaluate the generalizability of *MirageBeam*, we test its performance against two classic BFI-based WiFi sensing systems: BeamSense [7] and MuseFi [8]. As shown in Fig. 22, we measure both targeted and untargeted attack success rates across all action classes. *MirageBeam* achieves consistently high attack success on both systems, despite differences in model architectures and system implementations. For untargeted attacks, success rates remain above 80% in most cases. For targeted attacks, *MirageBeam* reaches over 70% accuracy in several action categories. These results demonstrate that our attack is transferable across diverse sensing pipelines.

E. Extension of Action Categories

To further examine behavior diversity, we expanded the original three-action setting to a five-action setting by adding two more refined actions, namely *kick* (K) and *circle* (C). The corresponding results are shown in Fig. 23. Both targeted and untargeted attacks remain consistently effective across all five categories. The success rates for both targeted and untargeted attacks remained above 70%, indicating that the proposed method

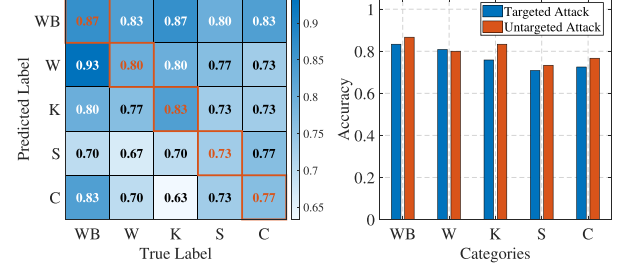


Fig. 23. Success rates of targeted and untargeted attacks across five expanded behavioral categories.

is not restricted to the small action set. The confusion matrix of the expanded action setting also supports the behavioral generality of the proposed attack.

F. Black-Box Attack

To evaluate the transferability of adversarial samples in black-box settings, we conduct cross-model attacks by generating perturbations using a surrogate model and applying them to unseen target models. Specifically, we first generate adversarial BFI sequences based on a 2-layer CNN surrogate. These samples are then directly applied to three black-box victim models: a 5-layer CNN, a GRU, and a Transformer. As shown in Table I, this naive transfer achieves moderate success on the 5-layer CNN and Transformer models. However, it performs poorly on the GRU model, achieving only 16.83% UASR and 10.80% TASR, highlighting the limitations of single-model-based transfer. To improve cross-model generalization, we introduce a model ensemble strategy [16] to fuse the logit outputs of the 2-layer CNN and BeamSense model by weighted averaging. The combined logits are used to construct a more robust fitness function during perturbation generation. This ensemble-based enhancement significantly boosts attack success across all victim models. These results confirm the effectiveness of ensemble-guided perturbation generation in enhancing black-box adversarial transferability. Future work may further enhance black-box attack performance through techniques such as GAN-based data augmentation and more comprehensive utilization of BFI.

VII. EFFECTIVENESS OF MIRAGEBEAM AGAINST POTENTIAL DEFENSES

We study the ability of common defense mechanisms to identify forged BFIs generated by *MirageBeam*, with detection implemented at three distinct levels: content, timing, and protocol.

A. Content-Level Defense

Advanced sensing systems may adopt similarity-based detection schemes to identify anomalous BFI sequences based on deviation from historical patterns. Let $x \in \mathbb{R}^{N_{\text{ele}}}$ denote a regular BFI sample and $x' \in \mathbb{R}^{N_{\text{ele}}}$ denote an adversarial (or suspicious) sample. We define the content distance

$$d(x, x') \triangleq \text{MSE}(x, x') = \frac{1}{N_{\text{ele}}} \|x - x'\|_2^2. \quad (21)$$

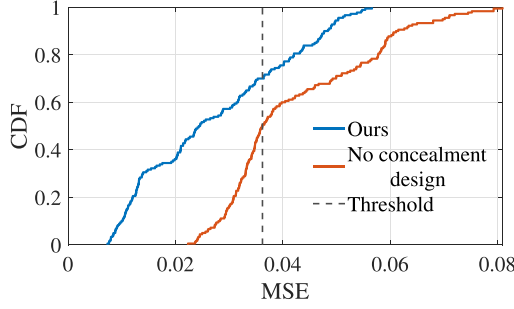


Fig. 24. The MSE distribution between the affected samples and the original samples.

From regular data, we estimate the mean and standard deviation of $d(\cdot, \cdot)$ as (μ, σ) and set the acceptance threshold

$$\tau \triangleq \mu + \sigma \quad (\mu = 0.0233, \sigma = 0.0129, \tau = 0.0362). \quad (22)$$

The content-level validator is

$$\mathcal{V}_{\text{cnt}}(x', x) = \mathbb{I}[d(x, x') \leq \tau], \quad (23)$$

where $\mathbb{I}[\cdot]$ is an indicator function. If the condition is met, the sample passes the content similarity check, then it equals 1; otherwise, it equals 0.

To counter this, *MirageBeam* explicitly incorporates a similarity-aware objective into its optimization process, minimizing the distance between forged and legitimate BFI signals in both time and subcarrier dimensions. This enables adversarial samples to maintain high statistical similarity with genuine traffic, effectively bypassing similarity-based anomaly detection. Fig. 24 presents the cumulative distribution function (CDF) of MSE values, demonstrating that the majority of our perturbed samples fall within a smaller error range. In general, the detection rate of the baseline reaches as high as 50.56%, while that of *MirageBeam* is significantly lower, at less than 30%.

B. Timing-Level Defense

Timing-level defenses detect forged BFIs by identifying abnormally dense injections through inter-arrival time analysis. Although regular BFI traffic is non-periodic and may be bursty under normal MAC scheduling, excessively short inter-arrival intervals typically indicate abnormal emission behavior. Accordingly, we focus on the left tail of the inter-arrival distribution. We compute BFI inter-arrival times from timestamps and plot empirical CDFs for two adversarial settings. As a regular reference, we use the 5th-percentile inter-arrival time ($P05$) measured under normal traffic as a lower-bound threshold.

As shown in Fig. 25, only a small fraction of samples (approximately 10%) fall below the regular $P05$ threshold across all settings, while the majority remain within the regular timing regime. These sub- $P05$ events may be considered short-term bursts in the MAC layer rather than sustained high-rate injection, indicating that the proposed attack largely preserves regular timing characteristics and evades rate-based timing validation.

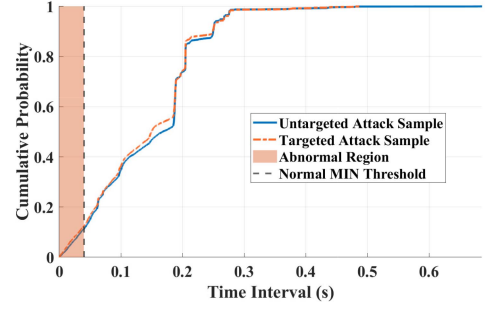


Fig. 25. The CDFs of BFI frame time intervals for different adversarial samples.

C. Protocol-Level Defense

Protocol-level defenses aim to reject forged BFIs by enforcing link- and MAC-layer consistency. Since BFIs are management frames generated within a well-defined sounding context, such defenses typically validate (i) transmitter identity, (ii) protocol semantics, and (iii) reception feasibility. We consider three representative mechanisms.

(D1) *Identity validation*: A BFI is accepted only if its source MAC address or BSSID matches a known legitimate transmitter.

(D2) *Protocol legitimacy validation*: BFIs must satisfy protocol semantics, such as plausible sequence numbers, dialog-token consistency within a sounding round, and consistency with a recent NDP-based sounding trigger.

(D3) *Decodability-based verification*: If a BFI is injected using the legitimate STA's identity, the legitimate STA is expected to overhear and successfully decode the frame; otherwise, the transmission is flagged as suspicious.

These defenses can be bypassed by a context-aware attacker. Specifically, **D1** is ineffective when forged BFIs reuse the legitimate STA's MAC identity. **D2** can be bypassed by passively observing the sounding context and injecting BFIs with protocol-plausible headers (e.g., valid tokens and sequence numbers), optionally marked as retransmissions to remain consistent with regular behavior. Finally, **D3** relies on the implicit assumption that forged BFIs using the STA's identity are decodable at the legitimate STA, which does not always hold in practice.

Bypassing D3 via asymmetric decodability: An attacker can enforce *asymmetric decodability*, where the forged BFI is decodable at the sensing receiver but not at the legitimate STA. Let S denote the sensing receiver and L denote the legitimate STA. For a forged frame transmitted with power P_t and modulation and coding scheme (MCS) k , successful decoding at receiver $X \in \{S, L\}$ requires

$$\Gamma_X(P_t) = P_t - \mathcal{L}(d_X) - \mathcal{N}_X \geq \theta_k, \quad (24)$$

where $\mathcal{L}(d_X)$ is the large-scale path loss, $\mathcal{N}_X$ denotes the effective noise-plus-interference level, and θ_k is the decoding threshold of MCS k . Asymmetric decodability is achieved when

$$\Gamma_S(P_t) \geq \theta_k + \frac{\varepsilon}{2}, \quad \Gamma_L(P_t) \leq \theta_k - \frac{\varepsilon}{2}, \quad (25)$$

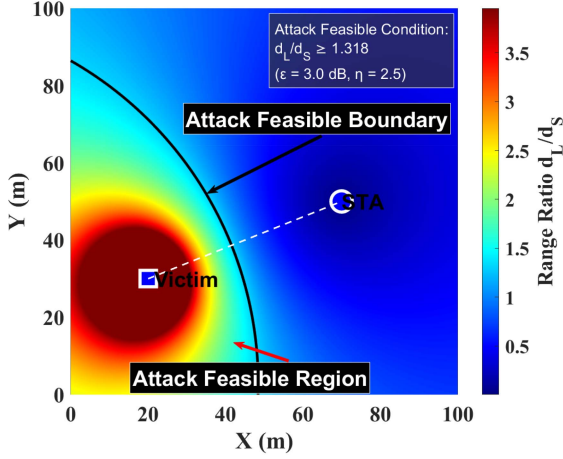


Fig. 26. Feasible attack region when SNR threshold ε is 3dB and η is 2.5.

for some robustness margin $\varepsilon > 0$.¹ We can further obtain

$$\mathcal{L}(d_L) - \mathcal{L}(d_S) \geq \varepsilon. \quad (26)$$

Assuming a log-distance attenuation model $\mathcal{L}(d) = \beta + 10\eta \log_{10}(d)$ (dB), where β denotes the propagation attenuation when d being 1m and η is the path loss exponent, (26) implies

$$\log_{10}\left(\frac{d_L}{d_S}\right) \geq \frac{\varepsilon}{10\eta}. \quad (27)$$

Fig. 26 visualizes the spatial feasibility of asymmetric decodability under a representative parameter setting. By appropriately selecting its spatial position and transmit power, the attacker can satisfy this condition, making the forged BFI decodable at the sensing receiver but undecodable at the legitimate STA. As a result, decodability-based verification fails, consistent with prior observations on asymmetric decodability [27].

VIII. DISCUSSIONS

This section explores enhancements to *MirageBeam* and prospective countermeasures to bolster BFI sensing security.

A. Label-Free Feasibility of Attacks

The attack evaluation in this work adopts labeled activity data to quantify attack effectiveness in a clear and reproducible manner. This setting is practical since in Section III-A we mentioned that attackers can not only obtain data labels through long-term observation, but also carry out attacks without relying on semantic activity labels.

To validate the feasibility of label-free attacks, we examine whether an attacker can infer the decision behavior of the target sensing model using unlabeled BFI samples only. Importantly, the attacker neither modifies nor attacks the unsupervised model. Instead, unsupervised learning is employed solely by the attacker as an auxiliary tool to organize unlabeled observations and derive surrogate decision indicators. Concretely, the attacker passively

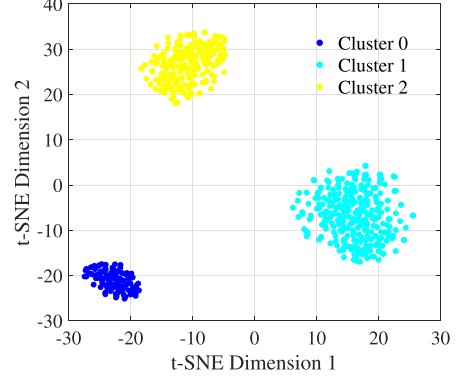


Fig. 27. Unsupervised clustering t-SNE results.

collects unlabeled BFI samples and applies an unsupervised learning pipeline consisting of representation learning followed by Deep Embedded Clustering (DEC). This auxiliary model is trained independently of the target classifier and does not require any semantic supervision. For the label-free setting, the reported DEC-based clustering result is obtained using 200 unlabeled passive BFI samples per action class, i.e., 600 samples in total for the three-action case. Each sample is assigned a cluster identifier, which serves as a non-semantic surrogate label for indexing attack objectives.

Fig. 27 visualizes the learned representations after DEC using t-SNE, and the samples are grouped into several well-separated clusters in the latent space. For evaluation purposes only, we perform an offline alignment between the inferred clusters and ground-truth activity labels. The resulting clustering accuracy reaches 83.4%, indicating that the internal decision structure of the sensing system can be reliably approximated through unsupervised observation alone. Such separation does not imply that the attacker manipulates the clustering boundary itself. Rather, it suggests that samples mapped to different clusters are likely to be separated by decision boundaries of the target classifier.

We further examined whether the learned cluster-to-label mapping remains stable across domains. The mapped clustering accuracy stays relatively high across different people (81.85% on person A and 80.24% on person B), but degrades more noticeably across days (79.2% to 66.11%) and environments (82.83% to 64.67%). This is consistent with the fact that cross-day and cross-environment shifts introduce stronger waveform variation in wireless sensing. We emphasize that this simple unsupervised pipeline is included to relax the attacker's label assumption, rather than to solve the more challenging problem of cross-domain unsupervised learning, which is beyond the scope of this work. However, this still proves that attackers can obtain possible semantic labels through unsupervised learning.

From an attack perspective, these surrogate groupings are sufficient to enable both untargeted and targeted attacks in a label-free manner. An untargeted attack can be realized by perturbing a sample such that the output of the target classifier changes, which in practice corresponds to crossing inferred surrogate decision regions. A targeted attack aims to steer the

¹Tables of MCS and associated SNR thresholds can be found on the following website: <https://wlanprofessionals.com/revolution-wifimcs-to-snr-levels/>.

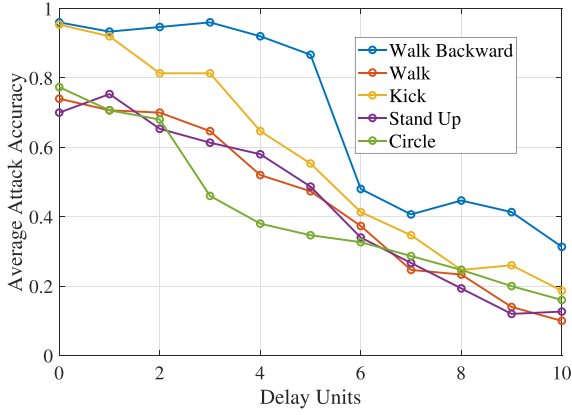


Fig. 28. Attack accuracy of different action types at various delay units.

target classifier's output from its original decision region into another specified region indexed by a chosen cluster identifier.

Overall, manipulating sensing outcomes at the decision level is sufficient to induce severe security consequences even under significantly weaker and more realistic attacker assumptions.

B. Delay Sensitivity Analysis

In practice, the onset of a human action is inherently ambiguous, and a small temporal neighborhood around the action start may all be considered reasonable segmentation points. To evaluate whether *MirageBeam* critically depends on exact synchronization, we further measure the average attack accuracy under different delay units, as shown in Fig. 28. The results show a gradual degradation trend rather than an abrupt collapse. Since both sample variability and delay mismatch are incorporated into the optimization process, *MirageBeam* does not rely on one exact trigger instant and remains effective under general small-to-moderate timing mismatch. When the delay becomes much larger, the attack effectiveness decreases more noticeably, because the forged BFI drifts farther away from the true action-start region.

C. Sensitivity to Candidate Timing Resolution L

The parameter L controls the temporal resolution of candidate injection timings within each behavior window. A larger L provides finer temporal flexibility and more candidate positions for placing forged BFI frames, whereas a smaller L restricts the attack space and may prevent the optimization from finding an effective targeted perturbation. To quantify this effect, we vary $L \in \{20, 30, 40, 48, 50\}$ and track, over iterations, the minimum implemented targeted loss L_t in (12) among all particles. A negative value of L_t means that the target-class logit already exceeds the maximum logit of all non-target classes, indicating that the targeted attack objective has been achieved. Therefore, the horizontal line at $L_t = 0$ in Fig. 29 serves as a natural decision boundary between unsuccessful and successful targeted optimization. Fig. 29 shows that when L is too small, the attack does not have sufficient temporal degrees of freedom, and the optimization cannot reliably push the minimum L_t below zero. For example, with $L = 20$, the targeted objective is difficult to

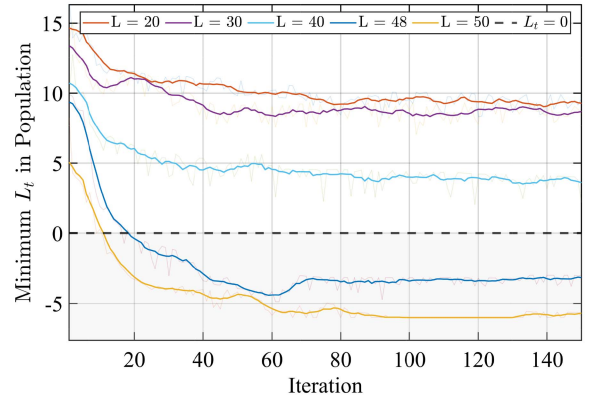


Fig. 29. Convergence behavior of minimum L_t

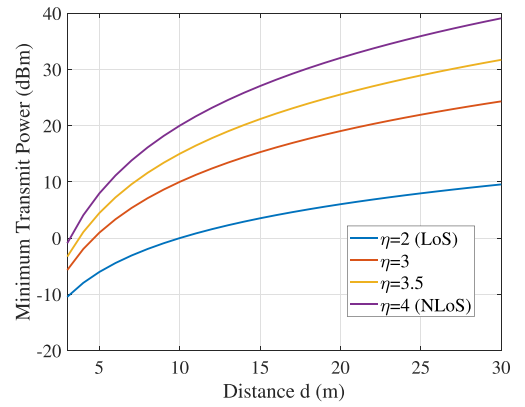


Fig. 30. Required transmit power under different distance and path loss exponents.

achieve consistently. As L increases to 30 and 40, the optimization becomes more effective, but convergence remains slower and less stable. When L reaches 48 and 50, the minimum L_t drops below zero more reliably, and the convergence behaviors become very similar. This indicates that the attack is clearly weakened only when L is too small, while it is not overly sensitive to the exact value of L once the candidate timing resolution is sufficiently fine. Therefore, the results suggest that $L = 50$ lies in a stable operating regime rather than being a narrowly tuned choice.

D. Feasibility Analysis in Different Environments

We further analyze the minimum transmit power required for a successful frame injection under different propagation conditions. The key to a successful attack lies in whether the forged frames can be successfully received by the victim, as discussed in (24). For a forged frame transmitted with power P_t and modulation and coding scheme (MCS) k , successful decoding at receiver X requires (6). For IEEE 802.11ac with MCS 6, typical parameters are $\theta_6 \approx 24\text{dB}$, $\beta \approx 46\text{dB}$, and $\mathcal{N}_X \approx -90\text{dBm}$. Typically, $\eta \in [2, 4]$ represents the path loss index for environments with different levels of complexity [28]. Fig. 30 illustrates the minimum transmit power required as a function of distance for different path loss exponents. Analysis

indicates that under NLoS conditions, the required transmission power increases due to higher attenuation; in other words, extremely complex environments can limit the success rate of attacks. However, within the transmission power limits of commercial Wi-Fi devices and at indoor distances of 5 to 10 meters, frame injection attacks remain feasible.

Similarly, the attacker–STA distance mainly affects whether the attacker can reliably overhear STA-related frames for onset triggering and timing alignment. When the distance is too great, *MirageBeam* will be unable to launch an effective attack. However, once the attacker can receive these frames reliably, the distance does not intrinsically change the crafted BFI content or the optimization objective, and thus does not materially affect attack success. Therefore, with commodity NICs and SDRs and 18–20 dBm transmit power, typical indoor ranges of 5–10 m allow reliable reception in our environments, making this factor non-limiting in our evaluation.

E. Mitigation

Beamforming Transmission of NDP: The NDP includes a predefined bit sequence known as the Long Training Field (LTF), which is known to the beamformer upon decoding. Since the primary function of the NDP is to probe the channel for channel state estimation, it is transmitted without beamforming by the access point. To counteract sensing-based attacks, an effective defense is to cryptographically obfuscate the NDP by applying a unitary transformation. Specifically, the LTF is pre-multiplied by a unitary matrix $\mathbf{Q}_k$, yielding: $\mathbf{H}_k \mathbf{Q}_k = \mathbf{U}_k \Sigma_k (\mathbf{Q}_k^H \mathbf{V}_k)^H$, which results in an obfuscated feedback at the STA: $\mathbf{V}'_k = (\mathbf{Q}_k^H \mathbf{V}_k)^H$. In this way, the legitimate user can still restore the normal BFI based on the known $\mathbf{Q}_k$, but the attacker can only obtain the time-varying obfuscated BFI, and thus cannot establish a mapping between the action pattern and the BFI sequence.

Encryption of BFI: One potential strategy is to encrypt BFI frames to prevent forged packets from bypassing security mechanisms. While this approach can significantly enhance security, it introduces considerable latency and overhead due to the frequent key exchange and the computational cost of encryption and decryption processes.

Adversarial Sample Identification: As a possible countermeasure, an anomaly-based detector can be introduced to identify suspicious BFI samples before they are processed by the sensing model. In our implementation, each sample is represented by eight simple statistical descriptors, including the global mean, global standard deviation, the mean, maximum, and standard deviation of adjacent-frame differences, the proportion of abrupt temporal changes, and the maximum and minimum values. Based on these features, an Isolation Forest is trained using only benign BFI samples to model the normal data distribution. Experimental results show that such a detector can identify 93.36% of the adversarial samples. However, it also misclassifies 21.07% of benign samples as adversarial. This indicates that, although this mitigation strategy can reduce the effectiveness of forged-frame attacks, it does so at the expense of normal system usability. Therefore, while anomaly-based filtering provides a

certain level of protection, it cannot fully resolve the security threat without introducing non-negligible service degradation.

Adversarial Training: To mitigate the impact of forged-frame adversarial samples, one possible solution is to incorporate labeled adversarial BFI samples into the training set and retrain the classifier. In our experiments, this adversarial training strategy resulted in a classification accuracy of 76.62% for adversarial examples. However, the benign-sample accuracy decreases by about 26.91%. This indicates that, although adversarial training can improve robustness against the proposed attack, it also introduces a substantial usability loss for normal sensing tasks. In addition, such a mitigation requires collecting and labeling adversarial samples in advance, which brings extra overhead to the defender. Therefore, this defense strategy still needs further improvement to better balance robustness and usability.

From a defense perspective, the above discussion suggests the representative protection directions. In practice, these stronger protections can be complemented by lightweight on-air checks, such as MAC/BSSID validation, sounding-context consistency checks, and reception-feasibility verification, to detect suspicious forged BFI frames before they are consumed by the sensing pipeline.

F. Limitations

Concurrent Multi-person Scenarios: Our threat model and evaluation focus on single-person activity sensing. Concurrent multi-person occupancy can yield overlapping motions and mixed BFI patterns, which may change feature distributions and make onset/timing triggers more ambiguous. Extending *MirageBeam* (and corresponding defenses) to robust multi-person settings requires addressing the underlying multi-person sensing/disentanglement problem, which is an active and challenging research direction and is left for future work.

Impact of BFI Reporting Frequency: BFI reporting frequency can influence both attack effectiveness and concealment, because it determines how densely legitimate BFI frames appear in the sensing sequence. In general, lower reporting frequencies may allow a relatively small number of injected forged frames to have a noticeable impact, whereas higher reporting frequencies may require more carefully optimized injections to achieve a comparable effect. However, in practice, BFI reporting frequency is not determined by traffic type alone. It is governed by the AP-chipset/firmware feedback algorithm and is further influenced by device placement, channel variation, and traffic generation. In particular, when the wireless channel varies more rapidly, denser BFI feedback may be triggered to help maintain communication performance. We also observed in our experiments that not all samples follow a simple monotonic relation in which heavier traffic always results in a higher BFI reporting rate. This suggests that the actual BFI frequency is jointly shaped by multiple interacting factors, rather than by traffic load alone. In our experiments, we emulate realistic usage by randomly interleaving different traffic types during data collection, rather than fixing one traffic pattern for a long period. As a result, the current evaluation already reflects mixed practical conditions to some extent. Nevertheless, isolating the effect of

reporting frequency as a single controlled factor would require a more systematic study under controlled AP feedback behavior, channel dynamics, and heterogeneous device placements. Therefore, characterizing this dependence more systematically and optimizing generation strategies under different feedback frequencies remain important directions for future work.

Extensions of Different Protocols: We note that the current experimental validation is conducted under the IEEE 802.11ac setting. Evaluation under 802.11ax/OFDMA and later protocol generations is outside the scope of the present study and remains an important direction for future work.

G. Responsible Disclosure and Compliance.

This work is intended for authorized security evaluation and defense improvement of WiFi sensing systems. Accordingly, we do not publicly release low-level packet-injection scripts that would directly enable on-air exploitation; only the methodological description and evaluation results necessary for scientific understanding are disclosed. All experiments were conducted in controlled indoor environments using authorized devices and infrastructure, and the study protocol was reviewed and approved by the ethics committee of our university.

IX. RELATED WORK

We elaborate on the related work of this paper from two aspects, WiFi sensing and adversarial attacks, and also clarify the differences between existing work and ours.

WiFi Sensing: WiFi sensing has been applied to gesture recognition [29], [30], [31], intrusion detection [32] and fall detection [33], [34], primarily based on CSI. However, CSI access is limited by hardware and driver constraints. BFI has recently emerged as a practical alternative, available on commodity devices without firmware modifications. BFI-based sensing has shown promise in tasks such as respiration monitoring [2], [3], activity recognition [17], and fine-grained motion analysis [8]. Meanwhile, WiFi sensing also introduces privacy risks, enabling inference of user location [35], [36], activity detection [37] and keystroke inference [4], [5], [38], [39].

Adversarial Attacks: Adversarial examples have been widely studied in vision [40], [41], [42] and speech [43], [44], [45], [46], with limited attention to WiFi sensing. In recent years, adversarial attacks have begun to target WiFi sensing systems [12], [47], [48]. Early WiFi sensing attacks [11], [12] operated on receiver-side data but lacked practical deployability. Later work introduced more practical approaches. Signal blocking [15] attempts to disrupt the sensing process by preventing frame reception, but its effectiveness is reduced in BFI systems that can recover missing data using sparse reconstruction, as demonstrated in MuseFi [8]. RF interference methods [13], [14] create additional reflection paths to perturb the subcarrier-level characteristics of CSI. Yet, BFI only contains compressed steering vectors and average gain, making it difficult to establish a reliable mapping between such perturbations and their effect on BFI. As a result, these methods lack validation and interpretability in the BFI context. Finally, pilot manipulation [10] requires access to the legitimate transmitter in order to modify training symbols, which

is typically infeasible for real-world adversaries. These constraints hinder their direct applicability to BFI sensing systems. To our knowledge, no prior work has demonstrated a practical attack on BFI-based sensing. In contrast, our work fills this gap by proposing *MirageBeam*, the first BFI-specific adversarial framework implemented entirely on commodity WiFi hardware.

X. CONCLUSION

In this paper, we reveal critical security vulnerabilities in BFI-based WiFi sensing via *MirageBeam*, the first practical adversarial framework for such systems. It demonstrates that forged BFI frames can reliably trigger untargeted and targeted misclassification in commodity behavior recognition systems, highlighting the urgency of securing this understudied domain. *MirageBeam* incorporates two key innovations to overcome core challenges: a joint optimization strategy that selects forged BFI amplitudes and injection timings to maximize attack efficacy while ensuring statistical indistinguishability from regular sequences, and a delayed-BFI injection window mechanism that enables seamless adaptation to real-world asynchrony caused by unpredictable injection delays. Extensive evaluations conducted across three distinct indoor environments validate the framework's effectiveness: it achieves 85.95% success in untargeted attacks and 81.81% in targeted attacks, while maintaining a detection rate of less than 30%. These results not only expose significant flaws in current BFI deployments but also provide actionable countermeasures, underscoring that the accessibility of BFI sensing must be paired with robust security mechanisms to foster the development of resilient WiFi technologies.

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